

Appendix IV



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Literature Review & Project Proposal for ST 4001 Senior Thesis

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Literature Review

1. Introduction

With the number of devices on the Internet of Things (IoT) growing exponentially, traditional centralized machine learning, where raw data is aggregated on a central server, faces critical bottlenecks in latency, bandwidth, and data privacy.[1] Federated Learning (FL) has emerged as a promising decentralized alternative, allowing devices to train models locally and share only model updates. However, in massive IoT networks, the communication overhead of transmitting data from thousands of devices remains a significant hurdle.[2]

To address this "communication wall," Over-the-Air Computation (AirComp) or Over-the-Air Federated Learning (OTA-FL) has been proposed. By exploiting the superposition property of the wireless multiple-access channel (MAC), OTA-FL allows simultaneous transmission of model updates, achieving bandwidth efficiency that is theoretically independent of the number of users.[3] However, this analog aggregation method exposes the learning process to impairments of the physical layer, specifically channels fading and non-Gaussian interference, which threaten the convergence and reliability of the global model. [4]

2. Current Research Fields

2.1 Research direction and current progress

Current research in OTA-FL is primarily focused on physical layer signal processing techniques to mitigate channel effects. The core principle relies on "channel inversion" power control, where transmitters pre-equalize their signals to ensure coherent superposition at the receiver.[3]

Recent breakthroughs include:

- **Beamforming and MIMO:** Utilization of multi-antenna systems at the edge server to enhance signal alignment and suppress interference through optimized receive beamforming.[5]
- **Reconfigurable Intelligent Surfaces (RIS):** Deployment of RIS to actively shape the wireless environment, mitigating deep fading and improving the signal-to-noise ratio (SNR) of the aggregated signal.[6]
- **Device Scheduling:** Algorithms to select a subset of devices with favorable channel conditions to participate in aggregation, thus reducing the error variance [7].

2.2 Existing problems and/or obstacles

Despite progress in signal alignment, fundamental gap between the statistical nature and industrial interference still remains.

- **Non-Gaussian Noise:** Most existing works assume Additive White Gaussian Noise (AWGN). However, measurement campaigns in industrial zones, e.g. factories, smart grids, consistently show that electromagnetic interference is impulsive and heavy-tailed, better modeled by Symmetric α -Stable distributions.[8]
- **Algorithmic Fragility:** Under heavy-tailed noise (where variance can be infinite), standard Stochastic Gradient Descent (SGD) algorithms used in FL are known to diverge. A single noise impulse during aggregation can corrupt the global model parameters.[9]
- **Lack of Adaptivity:** Current OTA-FL protocols typically employ static or pre-scheduled learning rates that do not adapt to the instantaneous volatility of the wireless channel.[4][10]

3. Prospective in the Research Field

The field is moving toward cross-layer designs that integrate algorithmic robustness with physical layer efficiency. There is a significant opportunity to explore adaptive optimization methods,

e.g. variants of Adam or AdaGrad, within the OTA-FL context.[11] In digital learning, these methods effectively handle sparse or noisy gradients by dynamically adjusting step sizes.[10] The research community speculates that adapting these mechanisms to handle physical channel noise—effectively using the optimizer as a noise filter—could enable robust learning in harsh industrial environments without the latency penalties of digital error correction.[9]

Project Proposal

1. Research Background

1.1 Introduction to the research fields

As Industrial IoT applications demand real-time intelligence, the deployment of FL over wireless edges is becoming critical. OTA-FL offers the necessary low latency for applications including autonomous driving and factory automation.[12] However, the industrial wireless environment is much more hostile than theory due to the impulsive noise from heavy machinery.[8] Existing OTA-FL solutions, which rely on Gaussian noise assumptions, are much likely to fail in these realistic deployment scenarios. This project focuses on the intersection of wireless communications (physical layer) and distributed optimization (algorithmic layer) to address this reliability challenge.

1.2 Aim and significance of the proposal

The primary aim of this project is to implement and validate an adaptive optimization framework for OTA-FL that functions as an implicit noise-mitigation filter. The project seeks to reproduce and extend some recent findings showing adaptive step-size mechanisms, e.g. Ada-Grad and Adam variants, can stabilize training trajectories against channel impairments.

The significance of this proposal includes:

- **Robustness:** It addresses the critical issue of heavy-tailed interference and fading in analog computing without requiring explicit channel inversion or perfect CSI.
- **Scalability:** By leveraging OTA computation, the system supports a massive amount of clients with constant communication latency.

- Efficiency: The proposed adaptive methods differ from standard approaches by utilizing second-order moments to dynamically compress the effective learning rate in the presence of high noise.

2. Proposed Research Approaches

2.1 Major Research Approaches

The project will follow the "simulation-based theoretical validation".

- Modeling: We will model the OTA-FL system with a "Noisy Oracle" that returns gradients corrupted by α -stable noise ($1 < \alpha \leq 2$).
- Algorithm Design: We propose to modify AdaGrad and Adam to use fractional moments (α -norms) rather than Euclidean norms for their second-moment estimation, hypothesizing that this will prevent the "gradient explosion" caused by noise impulses.[10][11]
- Analysis: We will attempt to derive the convergence rates for these modified algorithms, specifically looking for dependencies on the tail index α .

2.2 Techniques and methodology

- Simulation: A custom PyTorch environment will be built to simulate the analog superposition and noise injection. This avoids the overhead of full network simulators while capturing the essential mathematical properties of the physical layer for our research.[4][10]
- Noise Generation: We will use the Chambers-Mallows-Stuck method to generate synthetic α -stable noise with tunable tail indices.[8][13]
- Robust Statistics: We plan to integrate Median Anchored Clipping (MAC) as a pre-processing step to truncate extreme outliers before they actually reach the optimizer.[2]

2.3 Feasibility analysis

Adaptive gradient methods are well-established in the machine learning community for handling noisy datasets.[14] The extension to physical noise is conceptually sound. Previous literature review suggests that heavy-tailed noise can be managed if the learning rate decays sufficiently fast.[2] The necessary computational resources (GPUs for training ResNet models) are available.

3. Project Schedule and Expected Outcome

3.1 Research schedule

- Weeks 1-3: Develop "Noisy Oracle" system model and simulate fading channels/heavy-tailed noise.
- Week 4: Prepare datasets (MNIST, CIFAR-10) and basic model architectures.
- Weeks 5-10: Design and implement the adaptive mechanisms (AdaGrad-OTA, Adam-OTA). Conduct initial trials to tune hyperparameters.
- Weeks 11-14: Perform comparative analysis against baselines (FedAvg, FedAvgM) to verify hypotheses.
- Weeks 15-16: Conduct ablation studies (varying noise parameters α , number of clients N) and finalize the thesis.

3.2 Expected outcome

- Framework: A robust OTA-FL training framework capable of converging in impulsive noise environments where standard methods fail.

- **Theoretical Insight:** We expect to establish a theoretical link between the noise tail index α and the required learning rate decay for convergence. We hypothesize that Adam-OTA will show faster convergence than AdaGrad-OTA due to its momentum handling.
- **Validation:** Empirical evidence that shows ADOTA-FL achieves higher accuracy and lower loss than static baselines on standard benchmarks.

4. Research Progress

4.1 Preliminary results

We have currently completed the "System Model" phase (Phase 1) of the task plan.

- **Environment Setup:** We have successfully implemented a PyTorch-based simulation environment that emulates the analog OTA aggregation process. This "Noisy Oracle" is capable of injecting symmetric α -stable noise with tunable parameters.[10]
- **Prototype Implementation:** We have begun the coding structure for the adaptive optimizers. The initial code for estimating gradient moments on the server side has been written and is currently undergoing unit testing.

References

- [1] McMahan, H. B., Moore, E., Ramage, D. Arcas & y Arcas, B. A. Federated learning of deep networks using model averaging. *CoRR*, **2016**.
- [2] Li, J., Chen, Z., Chong, K. F. E., Das, B., Quek, T. Q. Yang & Yang, H. H. Robust federated learning over the air: Combating heavy-tailed noise with median anchored clipping. In *2025 23rd International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt)*. IEEE, **2025**, 89–96.
- [3] Zhu, G., Wang, Y. Huang & Huang, K. Low-Latency Broadband Analog Aggregation for Federated Edge Learning. *CoRR*, **2018**, abs/1812.11494, , <http://arxiv.org/abs/1812.11494>.
- [4] Yang, H. H., Chen, Z., Quek, T. Q. Poor & Poor, H. V. Revisiting analog over-the-air machine learning: The blessing and curse of interference. *IEEE Journal of Selected Topics in Signal Processing*, **2021**, 16(3), 406.
- [5] Zhu, G. Huang & Huang, K. MIMO over-the-air computation for high-mobility multi-modal sensing. *IEEE Internet of Things journal*, **2018**, 6(4), 6089.
- [6] Xiao, B., Yu, X., Ni, W., Wang, X. Poor & Poor, H. V. Over-the-air federated learning: Status quo, open challenges, and future directions. *Fundamental Research*, **2024**.
- [7] Amiri, M. M. Gündüz & Gündüz, D. Machine learning at the wireless edge: Distributed stochastic gradient descent over-the-air. *IEEE Transactions on Signal Processing*, **2020**, 68, 2155.
- [8] Clavier, L., Pedersen, T., Larrad, I., Lauridsen, M. Egan & Egan, M. Experimental evidence for heavy tailed interference in the IoT. *IEEE Communications Letters*, **2020**, 25(3), 692.
- [9] Chen, Z., Yang, H. H. Quek & Quek, T. Q. Edge intelligence over the air: Two faces of interference in federated learning. *IEEE Communications Magazine*, **2023**, 61(12), 62.
- [10] Wang, C., Chen, Z., Pappas, N., Yang, H. H., Quek, T. Q. Poor & Poor, H. V. Adaptive federated learning over the air. *IEEE Transactions on Signal Processing*, **2025**.
- [11] Duchi, J., Hazan, E. Singer & Singer, Y. Adaptive Subgradient Methods for Online Learning and Stochastic Optimization. *Journal of Machine Learning Research*, **2011**, 12, 2121.
- [12] Niknam, S., Dhillon, H. S. Reed & Reed, J. H. Federated learning for wireless communications: Motivation, opportunities, and challenges. *IEEE Communications Magazine*, **2020**, 58(6), 46.
- [13] Chambers, J., Mallows, C. Stuck & Stuck, B. A Method for Simulating Stable Random Variables. *Journal of The American Statistical Association - J AMER STATIST ASSN*, **1976**, 71, 340.
- [14] Kingma, D. P., Ba & Ba, J. Adam: A Method for Stochastic Optimization., **2017**, <https://arxiv.org/abs/1412.6980>.